

Android CameraX + Chaquopy + OpenCV

1. 專案初始化設定

同期中考第一題環境設定

2. Gradle 依賴

- Build.gradle(App)

```
dependencies {  
  
    // CameraX  
    def camerax = "1.3.2"  
    implementation "androidx.camera:camera-core:$camerax"  
    implementation "androidx.camera:camera-camera2:$camerax"  
    implementation "androidx.camera:camera-lifecycle:$camerax"  
    implementation "androidx.camera:camera-view:$camerax"  
    implementation "androidx.camera:camera-extensions:$camerax"  
  
}
```

3. Android 權限

- 在 AndroidManifest.xml 加入 : Camera 權限

```
<?xml version="1.0" encoding="utf-8"?>  
<manifest>  
  
    <uses-feature  
        android:name="android.hardware.camera"  
        android:required="true" />  
    <uses-permission android:name="android.permission.CAMERA"/>  
  
</manifest>
```

4. OpenCV Python 處理程式

- app/src/main/python/xxxx.py :

須留意輸入輸出資料格式轉換

```
import numpy as np
import cv2

def process_nv21(nv21_bytes: bytes, w: int, h: int) -> bytes:
    # NV21 → YUV420 → BGR
    yuv = np.frombuffer(nv21_bytes, dtype=np.uint8).reshape((h * 3 // 2, w))
    bgr = cv2.cvtColor(yuv, cv2.COLOR_YUV2BGR_NV21)

    # ===== 影像處理，示範 Canny =====
    gray = cv2.cvtColor(bgr, cv2.COLOR_BGR2GRAY)
    edge = cv2.Canny(gray, 80, 160)
    out = cv2.cvtColor(edge, cv2.COLOR_GRAY2BGR)
    # =====

    ok, buf = cv2.imencode(".png", out)
    if not ok:
        raise RuntimeError("encode failed")
    return buf.tobytes()
```

5. 版面配置 (activity_main.xml)

- PreviewView 顯示取景
- ImageView 疊在上層顯示結果
- Button 觸發快照

```
<!-- Camera 預覽 -->
<androidx.camera.view.PreviewView
    android:id="@+id/previewView"
    android:layout_width="0dp"
    android:layout_height="0dp"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toTopOf="@id/processBtn" />

<!-- 觸發按鈕 -->
<Button
    android:id="@+id/processBtn"
    android:text="Process"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginBottom="24dp"/>
```

```

<!-- Python 回傳的結果圖，預設隱藏 -->
<ImageView
    android:id="@+id/resultView"
    android:layout_width="0dp"
    android:layout_height="0dp"
    android:adjustViewBounds="true"
    app:layout_constraintDimensionRatio="16:9"
    android:scaleType="fitCenter"
    android:visibility="gone"
    app:layout_constraintTop_toTopOf="@id/previewView"
    app:layout_constraintBottom_toBottomOf="@id/previewView"
    app:layout_constraintStart_toStartOf="@id/previewView"
    app:layout_constraintEnd_toEndOf="@id/previewView"/>
<!--app:layout_constraintDimensionRatio="16:9"-->

```

6. CameraX 權限與初始

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    // 啟動 Chaquopy Python
    if (!Python.isStarted()) {
        Python.start(new AndroidPlatform(this));
    }
    py = Python.getInstance();

    setContentView(R.layout.activity_main);

    previewView = findViewById(R.id.previewView);
    resultView = findViewById(R.id.resultView);

    cameraExecutor = Executors.newSingleThreadExecutor();
    //startCamera(); // 啟動 CameraX

    Button btn = findViewById(R.id.processBtn);
    btn.setOnClickListener(v -> captureAndProcess());

    if (ContextCompat.checkSelfPermission(this, Manifest.permission.CAMERA)
        == PackageManager.PERMISSION_GRANTED) {
        startCamera();
    } else {
        cameraPermLauncher.launch(Manifest.permission.CAMERA);
    }
}

```

```

public class MainActivity extends AppCompatActivity {

```

```

private final ActivityResultLauncher<String> cameraPermLauncher =
    registerForActivityResult(new ActivityResultContracts.RequestPermission(),
        isGranted -> {
            if (isGranted) {
                startCamera();    // ← 僅在拿到權限後才綁定 CameraX
            } else {
                Toast.makeText(this, "需要攝影機權限才能運作", Toast.LENGTH_LONG).show();
            }
        });

```

- ProcessCameraProvider → Preview → ImageCapture

設定：

```

.setBufferFormat(ImageFormat.YUV_420_888)
.setTargetResolution(Size(640, 480))
.setCaptureMode(CAPTURE_MODE_MINIMIZE_LATENCY)

```

```

/** 初始化 CameraX : Preview + ImageCapture */
private void startCamera() {
    ListenableFuture<ProcessCameraProvider> future =
        ProcessCameraProvider.getInstance(this);

    future.addListener(() -> {
        try {
            ProcessCameraProvider cameraProvider = future.get();

            Preview preview = new Preview.Builder().build();
            preview.setSurfaceProvider(previewView.getSurfaceProvider());

            ImageCapture imageCapture = new ImageCapture.Builder()
                .setBufferFormat(ImageFormat.YUV_420_888)    // ★ 需要 CameraX 1.3+
                .setTargetResolution(new Size(640, 480))    // ← 新增
                .setCaptureMode(ImageCapture.CAPTURE_MODE_MINIMIZE_LATENCY)
                .build();

            CameraSelector selector = CameraSelector.DEFAULT_BACK_CAMERA;

            cameraProvider.unbindAll();
            cameraProvider.bindToLifecycle(
                this, selector, preview, imageCapture);

        } catch (Exception e) {
            Log.e("CameraX", "Failed to bind camera", e);
        }
    }, ContextCompat.getMainExecutor(this));
}

```

7.

擷取影像 → NV21

- ImageCapture.takePicture(... OnImageCapturedCallback)
- imageProxyToNv21() 正確處理 rowStride / pixelStride:

送進 Python

```
PyObject func = py.getModule("Hello").get("process_nv21");
PyObject result = func.call(nv21, width, height);
byte[] png = result.toJava(byte[].class);
```

更新 UI

- 在 UI Thread :
- Bitmap bmp = BitmapFactory.decodeByteArray(png, 0, png.length);
- resultView.setImageBitmap(bmp);
- resultView.bringToFront();
- resultView.setRotation(90f);

```
/** 按下按鈕擷取一幀 → Python 處理 → 回傳顯示 */
private void captureAndProcess() {
    imageCapture.takePicture(cameraExecutor,
        new ImageCapture.OnImageCapturedCallback() {

        @Override
        public void onCaptureSuccess(@NonNull ImageProxy image) {
            //Bitmap bmp = imageProxyToBitmap(image);
            byte[] nv21 = imageProxyToNv21(image);
            int width = image.getWidth();
            int height = image.getHeight();
            image.close();
        }

        cameraExecutor.execute(() -> {
            try {
                PyObject func = py.getModule("Hello").get("process_nv21");
                PyObject result = func.call(nv21, width, height);
                byte[] outPng = result.toJava(byte[].class);

                runOnUiThread(() -> {
                    Bitmap outBmp = BitmapFactory.decodeByteArray(outPng, 0, outPng.length);
                    resultView.setImageBitmap(outBmp);
                });
            } catch (Exception e) {
                e.printStackTrace();
            }
        });
    }
}
```

```

        resultView.setVisibility(View.VISIBLE);
        resultView.bringToFront();    // ← 關鍵，把它蓋到最上層
        resultView.setRotation(90f);
    });
} catch (Exception e) {
    Log.e("Python", "Error in Python call", e);
}
});
}

@Override
public void onError(@NonNull ImageCaptureException exc) {
    Log.e("CameraX", "Capture error", exc);
}
});
}
}

```

```

/** 正確處理 rowStride / pixelStride，把 YUV_420_888 轉成 NV21 */
private static byte[] imageProxyToNv21(ImageProxy image) {
    int w = image.getWidth();
    int h = image.getHeight();

    int ySize = w * h;
    int uvSize = w * h / 2;           // NV21: ¼ 像素數 ×2 色差 = ¼ Y
    byte[] nv21 = new byte[ySize + uvSize];

    // ----- Y -----
    ImageProxy.PlaneProxy yPlane = image.getPlanes()[0];
    ByteBuffer yBuf = yPlane.getBuffer();
    int yRowStride = yPlane.getRowStride();

    int pos = 0;
    for (int r = 0; r < h; r++) {
        yBuf.position(r * yRowStride);
        yBuf.get(nv21, pos, w);
        pos += w;
    }

    // ----- VU -----
    ImageProxy.PlaneProxy uPlane = image.getPlanes()[1];
    ImageProxy.PlaneProxy vPlane = image.getPlanes()[2];
    ByteBuffer uBuf = uPlane.getBuffer();
    ByteBuffer vBuf = vPlane.getBuffer();
    int rowStrideUV = uPlane.getRowStride();    // 兩平面相同
    int pixelStrideUV = uPlane.getPixelStride(); // 多數機型 = 2

    for (int r = 0; r < h / 2; r++) {
        int rowStart = r * rowStrideUV;
        for (int c = 0; c < w / 2; c++) {
            int uIdx = rowStart + c * pixelStrideUV;
            int vIdx = rowStart + c * pixelStrideUV;

            nv21[pos++] = vBuf.get(vIdx);    // ★ NV21: V 在前

```

```
        nv21[pos++] = uBuf.get(uIdx);    // ★          U 在後
    }
}
return nv21;
}
```

Camera X 主要功能如下，詳細功能可參考 [CameraX 總覽 | Android media | Android Developers](#)

Preview(圖像預覽): 看到的畫面設定(還沒要拍照存檔前的畫面)

Image analysis(圖像分析): 取得圖片的數據

Image capture(拍照): 拍下的照片設定，像是閃光

參考

[Android Camera2 入門系列 1 - Camera2 在 textureView 预览 - 簡書](#)

[Capture Image & Display in ImageView | Android App Development Tutorials | Part 1](#)



[\[Android\] 內建相機拍照實作\(1\) — 1010Code](#)

[How to Build a Camera App in Android Studio | Simple Camera App \(2022\)](#)

